

LEDGER · 2026-08-31 · AGRICULTURAL BIOTECHNOLOGY

Enzymatic Alginate Oligosaccharides (AOS) via Targeted Polysaccharide Lyase Depolymerization

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Acadian Plant Health Ascophyllum nodosum extract (AscoStar / Stella Maris)
Entry application	closed-system hydroponic and greenhouse operations
Measured advantage	Root biomass increase of ~0.97x to 1.1x relative to Ascophyllum nodosum extract (+40.0% AOS vs +41.1% ANE in wheat; +20.2% AOS vs +18% ANE in tomato/soybean)
Time to first revenue	4 months
Gross margin at parity	56.1%
Startup CapEx	\$265,000
Payback from first sale	101.5 months

Capital productivity

0.25x

Regulatory risk

Moderate — qualification costs reported (\$45,000 estimated)

Why it failed

- Voided by: Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression
- Below the 2x step-change bar

Assessment

The venture fails multiple key criteria, including gross margin (56.1% vs required 70%), initial CapEx (\$265,000 vs \$250,000 limit), and superiority delta (~0.97x to 1.1x vs required >2.0x). High input costs of refined sodium alginate and form factor regression further invalidate the concept.

Also flagged

- Price parity asserted, not sourced
- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-31 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.